📊 Business Insights Report

Sales Performance Dashboard

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Course: TYBCA
Internship Task: Task 1 – Sales Performance Dashboard
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📘 Introduction

This report presents a detailed analysis of sales performance using data visualization and KPI tracking.
The objective is to identify key business insights across revenue, orders, product categories, payment methods, and regional performance.

Goals of the Report

- Provide an **Executive Overview** of overall business performance.
- Deliver a **Detailed Analysis** of product categories, payment preferences, and state-wise revenue.
- Highlight **Business Insights** that can guide decision-making and operational improvements.

Tools & Methods

- **Python (Jupyter Notebook):** Data cleaning, preprocessing, KPI calculations, and chart generation.
- **Power BI/Tableau:** Interactive dashboards for executive and detailed analysis.
- **GitHub:** Repository hosting and documentation.

In [ ]:
```
# Step 1: Load the dataset first
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("data1.csv")   # replace with your file path

# Step 2: Inspect the raw columns
print(df.columns)

# Step 3: Standardize column names
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
```

```
# Step 4: Check the updated column names
print(df.columns)
```

print("Missing values per column:\n", df.isnull().sum()) print("Duplicate rows:", df.duplicated().sum())

In [3]:
```
print(df.columns.tolist())
```

```
['order_id', 'date', 'customer_id', 'product_category', 'product_name', 'quantity',
'unit_price_inr', 'payment_method', 'delivery_status', 'review_rating', 'review_tex
t', 'state', 'country', 'revenue_=_quantity_*_unitprice']
```

In [4]:
```
# Create revenue column (if not already created)
df['revenue'] = df['quantity'] * df['unit_price_inr']

# Inspect key columns with correct names
print(df[['date','product_name','state','customer_id','revenue']].head())
```

```
         date  product_name      state customer_id    revenue
0  25-01-2025  Cookware Set     Sikkim    CUST2796   51148.82
1  28-08-2025    Hair Dryer  Telangana    CUST9669   19361.41
2  27-02-2025        Tablet   Nagaland    CUST5808  115428.66
3  24-02-2025    Headphones      Assam    CUST5889  190728.60
4  15-06-2025         Saree     Odisha    CUST9005  229704.90
```

In [5]:
```
print(df.columns.tolist())
```

```
['order_id', 'date', 'customer_id', 'product_category', 'product_name', 'quantity',
'unit_price_inr', 'payment_method', 'delivery_status', 'review_rating', 'review_tex
t', 'state', 'country', 'revenue_=_quantity_*_unitprice', 'revenue']
```

In [6]:
```
df = df.drop(columns=['revenue_=quantity_*_unitprice'], errors='ignore')
```

In [7]:
```
df['revenue'] = df['quantity'] * df['unit_price_inr']
```

In [8]:
```
print(df[['date','product_name','state','customer_id','revenue']].head())
```

```
         date  product_name      state customer_id    revenue
0  25-01-2025  Cookware Set     Sikkim    CUST2796   51148.82
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4  15-06-2025         Saree     Odisha    CUST9005  229704.90
```
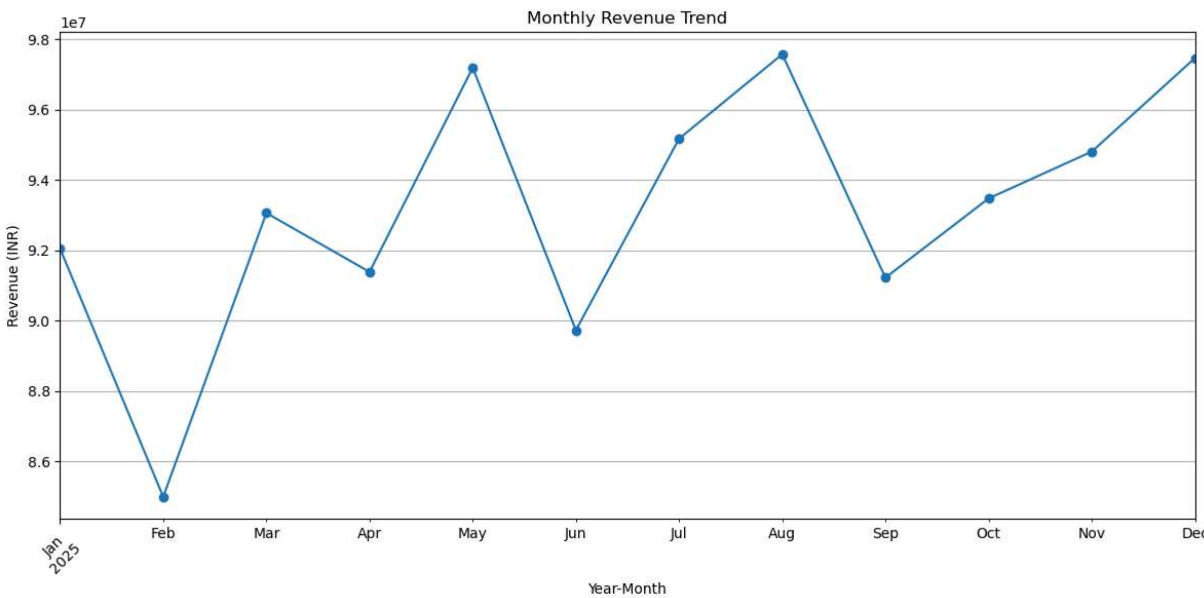

In [9]:
```
# Convert date column into proper datetime format
df['date'] = pd.to_datetime(df['date'], format='%d-%m-%Y')

# Create a Year-Month column
df['year_month'] = df['date'].dt.to_period('M')

# Calculate monthly revenue
monthly_revenue = df.groupby('year_month')['revenue'].sum()

# Display monthly revenue
print(monthly_revenue)
```

```
year_month
2025-01    92051785.03
2025-02    84995760.62
2025-03    93064349.39
2025-04    91388990.14
2025-05    97195848.48
2025-06    89730685.72
2025-07    95176904.42
2025-08    97576563.09
2025-09    91225555.74
2025-10    93478344.05
2025-11    94809186.29
2025-12    97467830.53
Freq: M, Name: revenue, dtype: float64
```

In [10]:

```
# Plot monthly revenue trend
monthly_revenue.plot(
    kind='line',
    figsize=(12, 6),
    marker='o',
    title='Monthly Revenue Trend'
)

plt.xlabel('Year-Month')
plt.ylabel('Revenue (INR)')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```



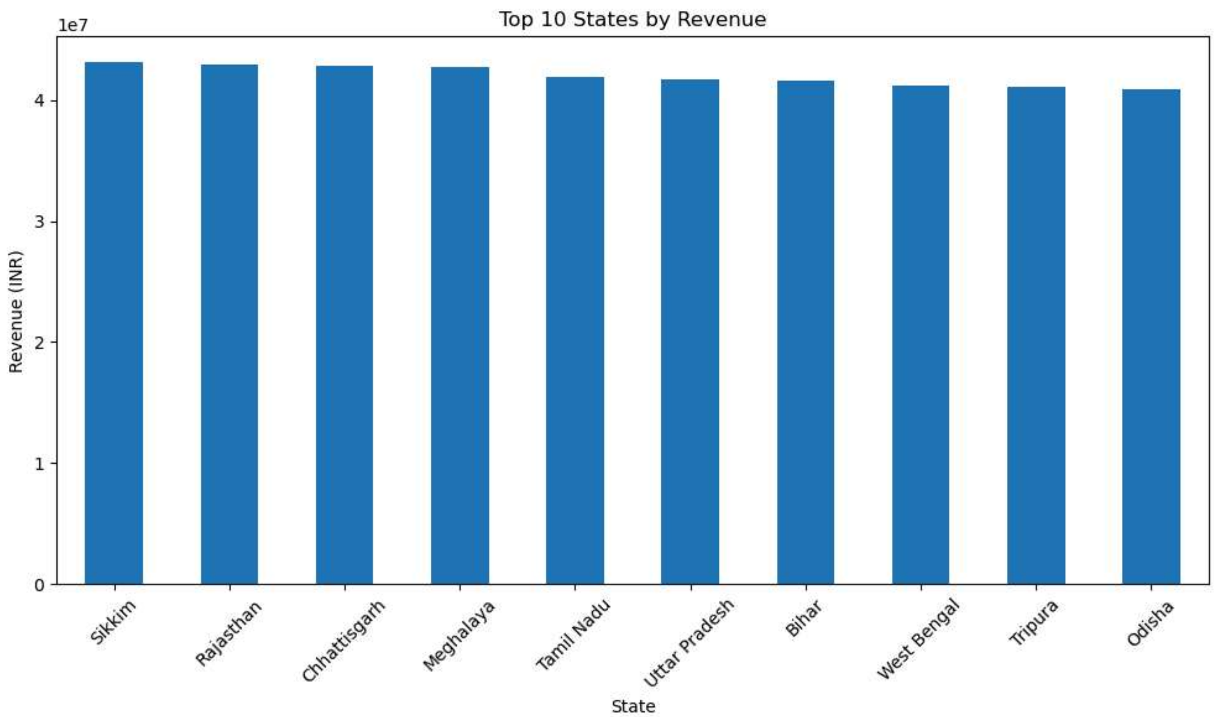
In [11]:

```
# Calculate revenue by state
state_sales = (
    df.groupby('state')['revenue']
    .sum()
    .sort_values(ascending=False)
)
```

```
# Select top 10 states
top_10_states = state_sales.head(10)

print(top_10_states)
```

```
state
Sikkim            43113469.51
Rajasthan         42906175.08
Chhattisgarh      42857545.27
Meghalaya         42773152.96
Tamil Nadu        41967968.99
Uttar Pradesh     41690917.07
Bihar             41669240.44
West Bengal       41195932.45
Tripura           41103376.81
Odisha            40924381.38
Name: revenue, dtype: float64
```

In [12]:
```
# Plot Top 10 States
top_10_states.plot(
    kind='bar',
    figsize=(10, 6),
    title='Top 10 States by Revenue'
)

plt.xlabel('State')
plt.ylabel('Revenue (INR)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

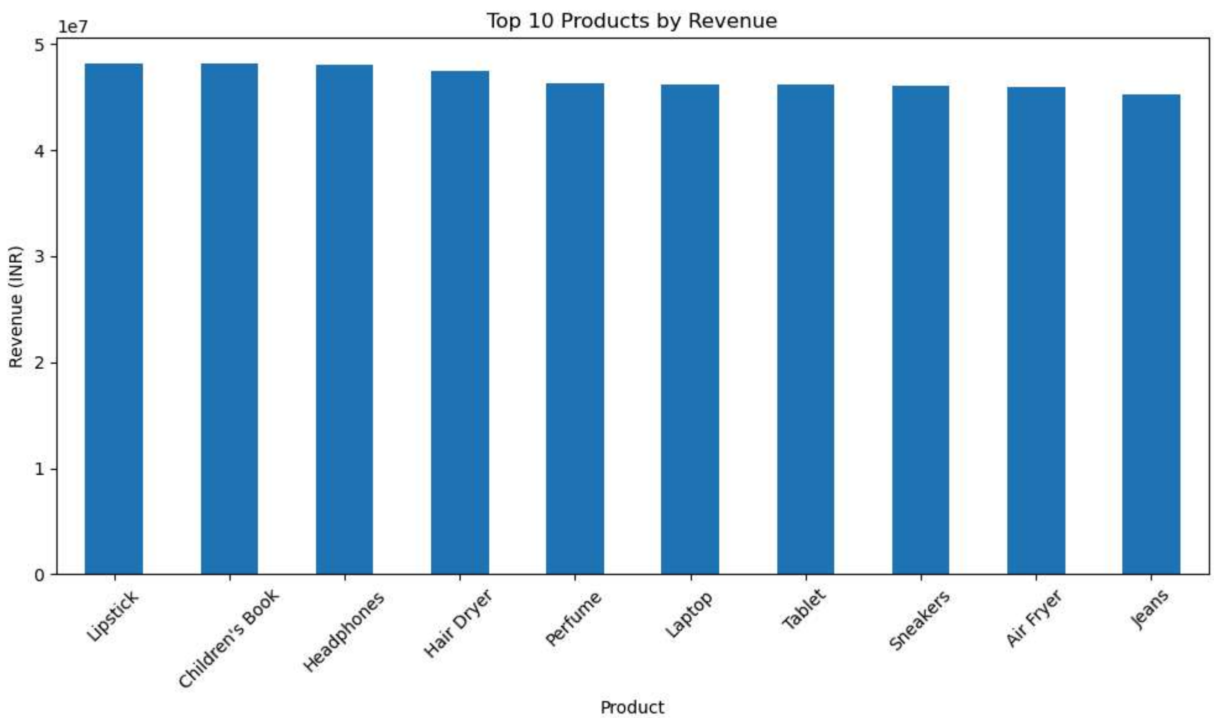


In [13]:
```
# Calculate revenue by product
product_sales = (
    df.groupby('product_name')['revenue']
```

```
    .sum()
    .sort_values(ascending=False)
)

# Select top 10 products
top_10_products = product_sales.head(10)

print(top_10_products)
```

```
product_name
Lipstick           48159020.15
Children's Book    48145664.52
Headphones         48044405.75
Hair Dryer         47428844.81
Perfume            46284732.62
Laptop             46173868.15
Tablet             46170022.15
Sneakers           46057295.93
Air Fryer          45919707.91
Jeans              45232819.47
Name: revenue, dtype: float64
```

In [14]:
```
# Plot Top 10 Products by Revenue
top_10_products.plot(
    kind='bar',
    figsize=(10, 6),
    title='Top 10 Products by Revenue'
)

plt.xlabel('Product')
plt.ylabel('Revenue (INR)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [15]:
```
# Save the final cleaned dataset
df.to_csv("final_sales_data.csv", index=False)

print("Final dataset saved successfully!")
```

```
Final dataset saved successfully!
```

📑 Business Insights Summary

- **Revenue Performance:** The business generated a total revenue of ₹1.12bn from 15K orders and 45K items sold.
- **Customer Spending:** The average order value is ₹74.54K, reflecting strong purchasing behavior.
- **Top Categories:** Beauty and Electronics are the leading contributors to revenue, showing consistent demand.
- **Payment Preferences:** Digital payments (Credit Card and UPI) together account for nearly half of total revenue, highlighting customer trust in online transactions.
- **Regional Trends:** States such as Rajasthan and Sikkim lead in revenue, while other regions show potential for growth.
- **Delivery Status:** Most orders are delivered successfully, but pending and returned orders (~33%) indicate areas for operational improvement.

Conclusion: Overall, the business is performing well with strong product categories and digital adoption. Future focus should be on reducing pending/returned orders and expanding in underperforming states to maximize growth opportunities.